

Tutorato di algoritmi e laboratorio 03

Oggi parleremo di counting-sort...

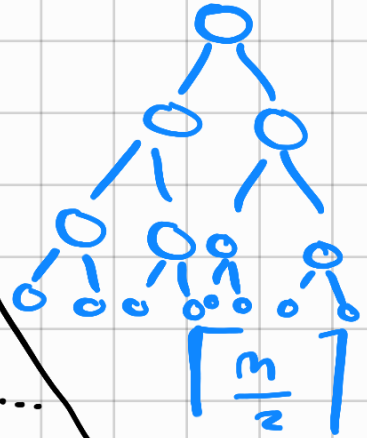
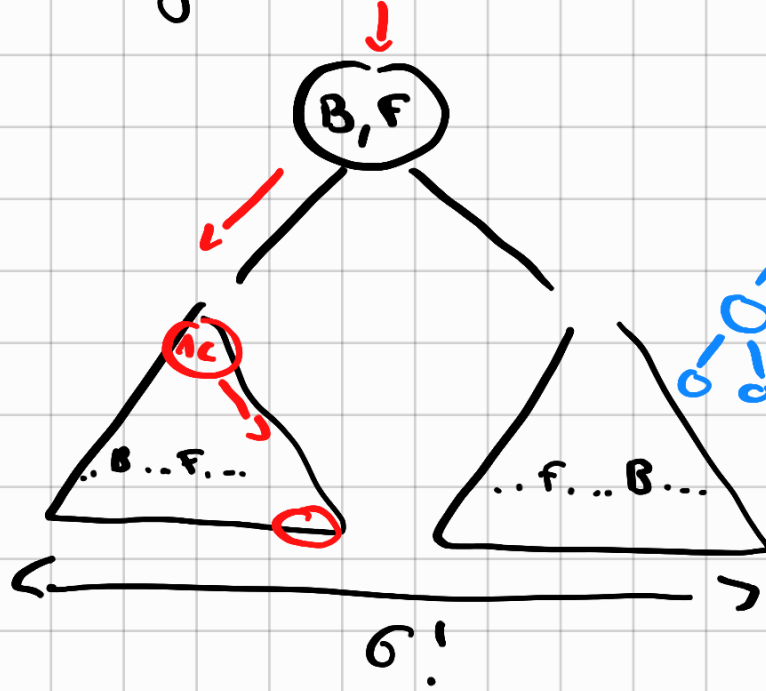
Domanda... E' veramente l'algoritmo migliore?

Scannatevi

Il limite dell'ordinamento per confronti...

$$V = \langle A, C, B, F, E, D \rangle$$

Domanda che informazione ci dà un confronto?



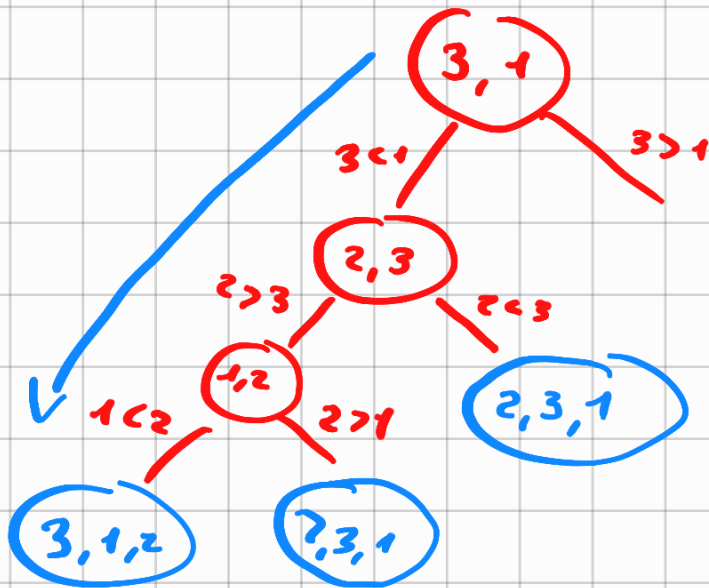
$$h = \log_2 n$$

$$f = \sqrt{\frac{2}{2}}$$

$$m \approx 28$$

$$h = \lg_2 z = f - \lg_2 d-1$$

3, 1, 2

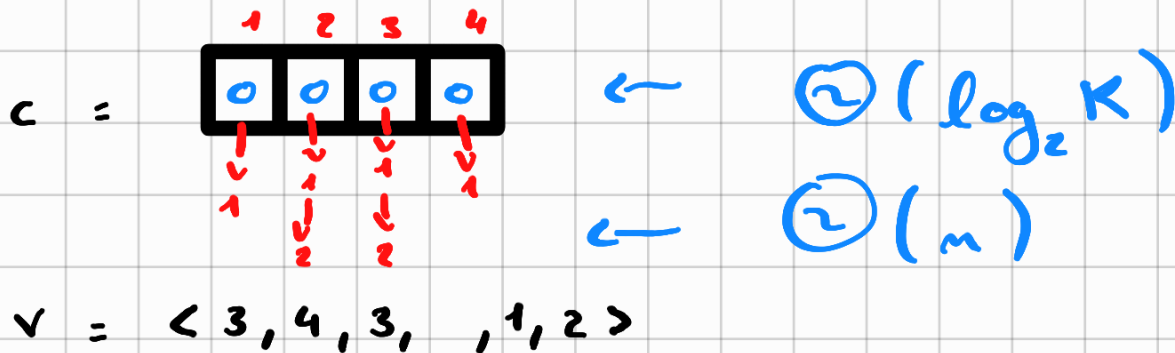


Counting sort

Range = $\{1, 2, 3, 4\}$

Vettore $v = \langle 3, 4, 3, 2, 1, 2 \rangle$

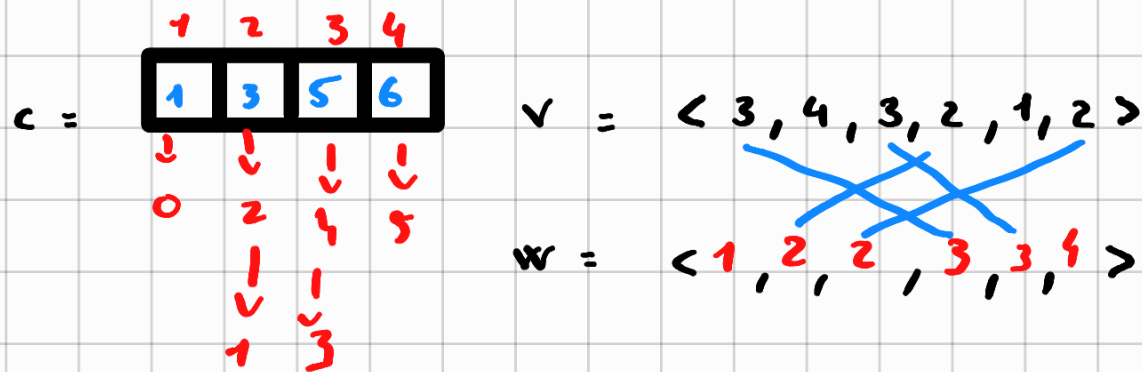
Passo 1)



Passo 2)



Passo 3)



Proviamo a creare una pseudocodice?

counting-sort ($v = \langle v_1, \dots, v_n \rangle, n$)

$\Theta(n)$ { $M \leftarrow \max(v, n)$

+

$\Theta(n)$ { $m \leftarrow \min(v, n)$

+

$\Theta(1)$ $r \leftarrow M - m + 1$

+

$\Theta(1)$ $c \leftarrow$ array di lunghezza r inizializzato a 0
// $c \in \mathbb{N}^r : c_i = 0 \forall i: 1 \leq i \leq r$

+

$\Theta(n)$ for $i = 1$ up to n do

+

$\Theta(1)$ $c[v[i] - m + 1] \leftarrow c[v[i] - m + 1] + 1$

$\Theta(n)$ for $i = 1$ up to $r - 1$ do

+

$\Theta(1)$ $c[i + 1] \leftarrow c[i] + c[i + 1]$

$\Theta(n)$ $w \leftarrow$ array di lunghezza n

+

$\Theta(n)$ for $i = n$ down to 1 do

$\Theta(1)$ $w[c[v[i] - m + 1]] \leftarrow v[i] - m + 1$

$\Theta(1)$ $c[v[i] - m + 1] \leftarrow c[v[i] - m + 1] - 1$
return w

Esercizio

$$v \in \mathbb{Z}^8 \quad v = \langle -2, 7, 10, -4, 0, 2, 3, -1 \rangle$$

PASSO 1) $\text{MAX} = 10$ $\text{MIN} = -4$

-4, -3, -2, -1, 0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10

1 2 3 4 5 7 7 10 13 14 15

1 0 1 1 1 0 1 1 0 0 0 1 0 0 1

1 1 2 3 4 4 5 6 6 6 6 7 7 7 8



Avete sentito parlare di Radix Sort ?

Come può essere utile per gli informatici ?

Esercizi

Max-Heap

$v = \langle 2, 7, 3, 1, 8, 0, 9, 10, 5, 4, 11, 12 \rangle$

$H \leftarrow \text{Max-Heapify}(v)$

$\text{Extract-Max}(H)$

$\text{Extract-Max}(H)$

$\text{Extract-Max}(H)$

$\text{Extract-Max}(H)$

$\text{Insert}(H, 15)$

$\text{Insert}(H, 14)$

$\text{Insert}(H, 13)$

$\text{Extract-Max}(H)$

$\text{Extract-Max}(H)$

$\text{Extract-Max}(H)$

$\text{Extract-Max}(H)$